Introduction to Statistical Investigations

Second Edition

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Chapter 1.5

Significance: How Strong is the Evidence?

Inference for a Single Proportion: Theory-Based Approach

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Simulation-Based vs. Theory-Based

- We will now look at the more traditional method of determining a *p*-value through theory-based techniques.
- When we used simulation-based methods, we all got slightly different *p*-values. The more repetitions we would do, the closer our *p*-values were to each other.
- In theory-based methods, we will use a theoretical distribution to model our null distribution and we will all get the same *p*-value.

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Theory-Based Techniques

- Hopefully, you've noticed the shape of most of our simulated null distributions were quite predictable.
- We can predict this shape using normal distributions.
- When we do a test of significance using theory-based methods, only how our *p*-values are found will change. Everything else will stay the same.

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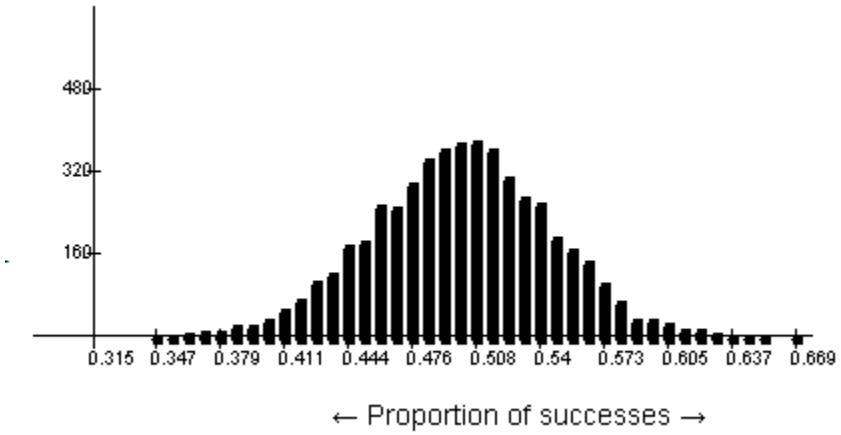
The Null Distributions

- Our null distributions:
  - Were typically bell-shaped
  - Centered at the proportion under the null
  - Width was dependent mostly on the sample size.



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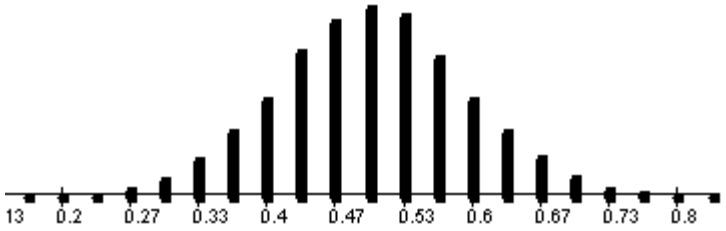
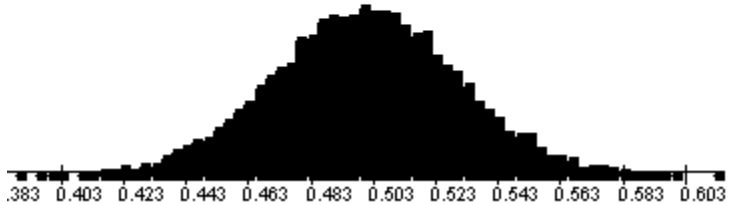
The Normal Distribution

- Both of these are centered at 0.50.
  - The one on the left represents samples of size 30.
  - The one on the right represents samples of size 300.
  - Both could be predicted using normal distributions.





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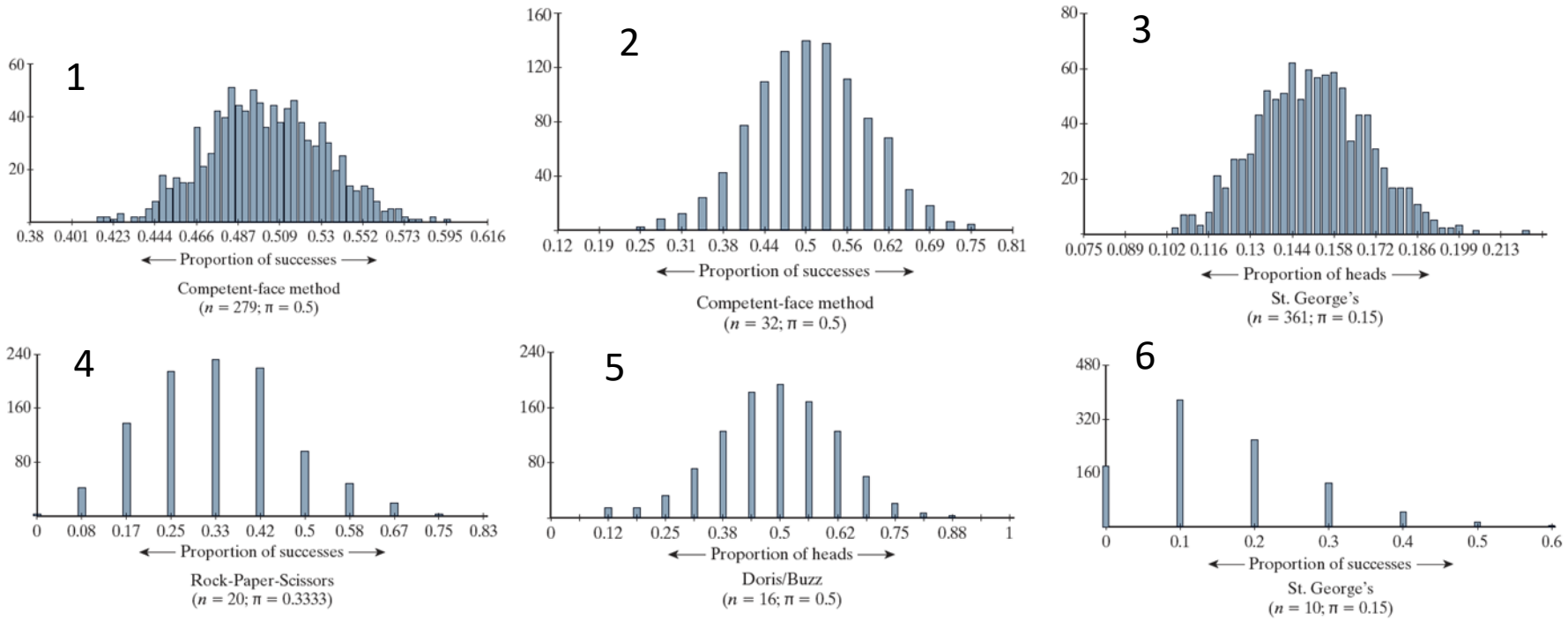
Examples from this Chapter

- Which ones will normal distributions fit?



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Advantages and Disadvantages of Theory-Based Tests

- **Advantages of theory-based tests**
  - No need to set up some simulation method
  - Fast and easy
  - Can be done with a wide variety of software
  - We all get the same *p*-value.
  - Determining confidence intervals (we will do this in Chapter 3) is much easier.
- **Disadvantages of theory-based tests**
  - They will all come with some validity conditions (like the number of successes and failures we have for a single proportion test).

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When Can I Use a Theory-Based Test that Uses the Normal Distribution?

- You should have at least 10 successes and 10 failures in your sample to be fairly confident that a normal distribution will fit the simulated null distribution nicely.

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Senate Elections

- 279 races
- 189 successes (predicted correctly)
- 279-189 = 90 failures (predicted incorrectly)
- Meets the validity condition?

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Example 1.5: Predicting Standard Deviation

$\sqrt{\dfrac{\pi(1-\pi)}{n}}$ where π is the proportion under the null (hypothesis), and n is the sample size.

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Example 1.5: Central Limit Theorem

The **Central Limit Theorem**.

If the sample size is *large enough*:

- The distribution of sample proportions will be bell-shaped (or normal)
- Centered at the long-run proportion (π)
- With a standard deviation of

$$\sqrt{\frac{\pi(1-\pi)}{n}}.$$

This means we can predict the shape, center, and variability of a null distribution and don't need to simulate under certain conditions (at least 10 successes and failures in our sample).

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Do I need to simulate? Or can I use the CLT?

Rock Paper Scissors:

- 20 games
- 6 games, someone uses scissors
- 20-6=14 games someone does not use scissors

Harley

- 10 tries
- 9 times he goes to the correct cup
- 1 time he doesn't go to the correct cup

Halloween Candy:

- 283 kids observed
  - 148 (52.3%) chose candy
  - 135 (47.7%) chose toys

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Do I need to simulate? Or can I use the CLT?

- **When validity conditions are not met, use simulation-based methods.**

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Example if time

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Example 1.5: Halloween Treats (1 of 4)

- Researchers investigated whether children show a preference for toys or candy
- Test households in five Connecticut neighborhoods offered children two plates:
  - One with candy
  - One with small, inexpensive toys
- The researchers observed the selections of 283 trick-or-treaters between ages 3 and 14.

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Example 1.5: Halloween Treats (2 of 4)

- Null: The long run proportion of trick-or-treaters who choose candy is 0.50.
- Alternative: The long run proportion of trick-or-treaters who choose candy is not 0.50.

- $H_0: \pi = 0.50$
- $H_a: \pi \neq 0.50$

Notice we are focusing on candy but could have easily done this focusing on the toy.

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Example 1.5: Halloween Treats (3 of 4)

- 283 children were observed
  - 148 (52.3%) chose candy
  - 135 (47.7%) chose toys
- Run the applet:
  - What is the p-value (two-sided)?
  - What is the standard deviation?

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- What is the standard deviation using the Central Limit Theorem?
- What is the z-statistic, using that computation?

$$z = \frac{(\text{observed statistic}) - (\text{mean of null distribution})}{(\text{SD of null distribution})}$$

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Example 1.5: Simulation- and Theory-Based Results from the One Proportion Applet

Probability of heads: 0.5

Number of tosses: 283

Number of repetitions: 1000

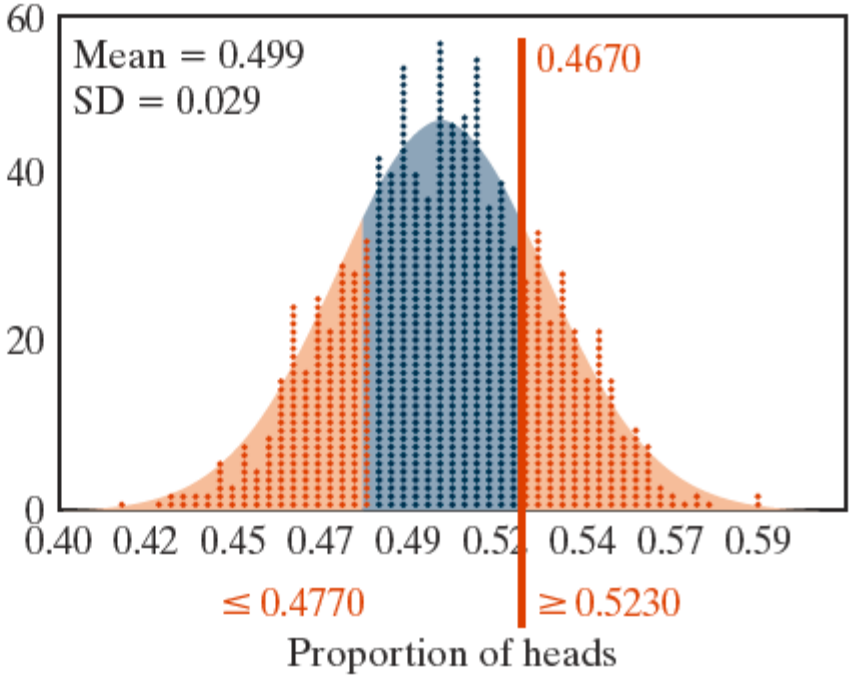
○ Number of heads
◉ Proportion of heads

As extreme as ≥ .523

Proportion of repetitions:
(233 + 234)/1000 = 0.4670

☑ Two-sided
☐ Exact Binomial
☑ Normal Approximation

p-value = 0.4390 (Z = 0.77)



$$\sqrt{\frac{0.5(1-0.5)}{283}} = 0.0297.$$

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Example 1.5: Halloween Treats (4 of 4)

- The theory-based *p*-value is 0.4390 so if half of the population of trick-or-treaters preferred candy, then there's a 43.90% chance that a random sample of 283 trick-or-treaters would have 148 or more, or 135 or fewer, choose the candy.
- Because it's not a small *p*-value, **we don't have strong (or even moderate) evidence that trick-or-treaters prefer one type of treat over the other.**

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